

Операционные системы

Анализ файловой структуры UNIX. Команды для работы с файлами и каталогами

Арсланов Довлетгелди

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Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

- 1 Выполнить приимеры
- 2 Выполнить дествия по работе с каталогами и файлами
- 3 Выполнить действия с правами доступа
- 4 Получить дополнительные сведения при помощи справки по командам.

2. Процесс выполнения лабораторной работы

```
darslanov@darslanov:~$ touch abc1
darslanov@darslanov:~$ cp abc1 april
darslanov@darslanov:~$ cp abc1 may
darslanov@darslanov:~$ mkdir monthly
darslanov@darslanov:~$ cp april may monthly
darslanov@darslanov:~$ cp monthly/may monthly/june
darslanov@darslanov:~$ ls monthly
april  june  may
darslanov@darslanov:~$ mkdir monthly.00
darslanov@darslanov:~$ cp -r monthly monthly.00
darslanov@darslanov:~$ cp -r monthly.00 /tmp
darslanov@darslanov:~$
```

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Рисунок 1: Выполнение примеров

```
darslanov@darslanov:~$  
darslanov@darslanov:~$ mv april july  
darslanov@darslanov:~$ mv july monthly.00  
darslanov@darslanov:~$ ls monthly.00  
july  monthly  
darslanov@darslanov:~$ mv monthly.00 monthly.01  
darslanov@darslanov:~$ mkdir reports  
darslanov@darslanov:~$ mv monthly.01 reports  
darslanov@darslanov:~$ mv reports/monthly.01 reports/monthly  
darslanov@darslanov:~$
```

Рисунок 2: Выполнение примеров

```
darslanov@darslanov:~$  
darslanov@darslanov:~$  
darslanov@darslanov:~$ touch may  
darslanov@darslanov:~$ ls -l may  
-rw-r--r--. 1 darslanov darslanov 0 Sep  4 12:29 may  
darslanov@darslanov:~$ chmod u+x may  
darslanov@darslanov:~$ ls -l may  
-rwxr--r--. 1 darslanov darslanov 0 Sep  4 12:29 may  
darslanov@darslanov:~$ chmod u-x may  
darslanov@darslanov:~$ ls -l may  
-rw-r--r--. 1 darslanov darslanov 0 Sep  4 12:29 may  
darslanov@darslanov:~$ chmod g-r,o-r monthly  
darslanov@darslanov:~$ chmod g+w abc1  
darslanov@darslanov:~$
```

Рисунок 3: Выполнение примеров

Создание директорий и копирование файлов

```
darslanov@darslanov:~$ cp /usr/include/linux/sysinfo.h ~
darslanov@darslanov:~$ mv sysinfo.h equipment
darslanov@darslanov:~$ mkdir ski.plases
darslanov@darslanov:~$ mv equipment ski.plases/
darslanov@darslanov:~$ mv ski.plases/equipment ski.plases/equiplist
darslanov@darslanov:~$ touch abc1
darslanov@darslanov:~$ cp abc1 ski.plases/equiplist2
darslanov@darslanov:~$ cd ski.plases/
darslanov@darslanov:~/ski.plases$ mkdir equipment
darslanov@darslanov:~/ski.plases$ mv equiplist equipment/
darslanov@darslanov:~/ski.plases$ mv equiplist2 equipment/
darslanov@darslanov:~/ski.plases$ cd
darslanov@darslanov:~$ mkdir newdir
darslanov@darslanov:~$ mv newdir ski.plases/
darslanov@darslanov:~$ mv ski.plases/newdir/ ski.plases/plans
darslanov@darslanov:~$
```

Рисунок 4: Работа с каталогами

Работа с командой chmod

```
darslanov@darslanov:~$ mkdir australia play
darslanov@darslanov:~$ touch my_os feathers
darslanov@darslanov:~$ chmod 744 australia/
darslanov@darslanov:~$ chmod 711 play/
darslanov@darslanov:~$ chmod 544 my_os
darslanov@darslanov:~$ chmod 664 feathers
darslanov@darslanov:~$ ls -l
total 0
-rw-rw-r--. 1 darslanov darslanov 0 Sep  4 12:34 abc1
drwxr--r--. 1 darslanov darslanov 0 Sep  4 12:35 australia
drwxr-xr-x. 1 darslanov darslanov 0 Sep  4 11:11 Desktop
drwxr-xr-x. 1 darslanov darslanov 0 Sep  4 11:11 Documents
drwxr-xr-x. 1 darslanov darslanov 0 Sep  4 11:11 Downloads
-rw-rw-r--. 1 darslanov darslanov 0 Sep  4 12:35 feathers
drwxr-xr-x. 1 darslanov darslanov 74 Sep  4 11:42 git-extended
-rw-r--r--. 1 darslanov darslanov 0 Sep  4 12:29 may
drwx--x--x. 1 darslanov darslanov 24 Sep  4 12:28 monthly
drwxr-xr-x. 1 darslanov darslanov 0 Sep  4 11:11 Music
-r-xr--r--. 1 darslanov darslanov 0 Sep  4 12:35 my_os
drwxr-xr-x. 1 darslanov darslanov 0 Sep  4 11:11 Pictures
drwx--x--x. 1 darslanov darslanov 0 Sep  4 12:35 play
drwxr-xr-x. 1 darslanov darslanov 0 Sep  4 11:11 Public
drwxr-xr-x. 1 darslanov darslanov 14 Sep  4 12:29 reports
drwxr-xr-x. 1 darslanov darslanov 28 Sep  4 12:35 ski.places
drwxr-xr-x. 1 darslanov darslanov 0 Sep  4 11:11 Templates
drwxr-xr-x. 1 darslanov darslanov 0 Sep  4 11:11 Videos
drwxr-xr-x. 1 darslanov darslanov 10 Sep  4 11:19 work
darslanov@darslanov:~$
```

```
root:x:0:0:Super User:/root:/bin/bash
bin:x:1:1:bin:/bin:/usr/sbin/nologin
daemon:x:2:2:daemon:/sbin:/usr/sbin/nologin
adm:x:3:4:adm:/var/adm:/usr/sbin/nologin
lp:x:4:7:lp:/var/spool/lpd:/usr/sbin/nologin
sync:x:5:0:sync:/sbin:/bin/sync
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown
halt:x:7:0:halt:/sbin:/sbin/halt
mail:x:8:12:mail:/var/spool/mail:/usr/sbin/nologin
operator:x:11:0:operator:/root:/usr/sbin/nologin
games:x:12:100:games:/usr/games:/usr/sbin/nologin
ftp:x:14:50:FTP User:/var/ftp:/usr/sbin/nologin
nobody:x:65534:65534:Kernel Overflow User:/usr/sbin/nologin
dbus:x:81:81:System Message Bus:/usr/sbin/nologin
tss:x:59:59:Account used for TPM access:/usr/sbin/nologin
systemd-oom:x:999:999:systemd Userspace OOM Killer:/usr/sbin/nologin
polkitd:x:114:114>User for polkitd:/sbin/nologin
systemd-coredump:x:998:998:systemd Core Dumper:/usr/sbin/nologin
systemd-timesync:x:997:997:systemd Time Synchronization:/usr/sbin/nologin
chrony:x:996:996:chrony system user:/var/lib/chrony:/sbin/nologin
systemd-network:x:192:192:systemd Network Management:/usr/sbin/nologin
systemd-resolve:x:193:193:systemd Resolver:/usr/sbin/nologin
avahi:x:70:70:Avahi mDNS/DNS-SD Stack:/var/run/avahi-daemon:/usr/sbin/nologin
unbound:x:993:993:Unbound DNS resolver:/var/lib/unbound:/sbin/nologin
clevis:x:992:992:Clevis Decryption Framework unprivileged user:/var/cache/clevis:/usr/sbin/nologin
```

Рисунок 6: Файл /etc/passwd

```
darslanov@darslanov:~$  
darslanov@darslanov:~$ cp feathers file.old  
darslanov@darslanov:~$ mv file.old play/ I  
darslanov@darslanov:~$ mkdir fun  
darslanov@darslanov:~$ cp -R play/ fun/  
darslanov@darslanov:~$ mv fun/ play/games  
darslanov@darslanov:~$ chmod u-r feathers  
darslanov@darslanov:~$ cat feathers  
cat: feathers: Permission denied  
darslanov@darslanov:~$ cp feathers feathers2  
cp: cannot open 'feathers' for reading: Permission denied  
darslanov@darslanov:~$ chmod u+r feathers  
darslanov@darslanov:~$ chmod u-x play/  
darslanov@darslanov:~$ cd play/  
bash: cd: play/: Permission denied  
darslanov@darslanov:~$ chmod +x play/  
darslanov@darslanov:~$
```

Рисунок 7: Работа с файлами и правами доступа

```
MOUNT(8)                                     System Administration                                     MOUNT(8)
```

NAME

mount - mount a filesystem

SYNOPSIS

mount [-h|-V]

mount [-l] [-t fstype]

mount -a [-fFnrsvw] [-t fstype] [-O optlist]

mount [-fnrsvw] [-o options] device | mountpoint

mount [-fnrsvw] [-t fstype] [-o options] device mountpoint

mount --bind|--rbind|--move olddir newdir

mount --make-[shared|slave|private|unbindable|rshared|rslave|rprivate|runbindable] mountpoint

DESCRIPTION

All files accessible in a Unix system are arranged in one big tree, the file hierarchy, rooted at /. These files can be spread out over several devices. The **mount** command serves to attach the filesystem found on some device to the big file tree. Conversely, the **umount(8)** command will detach it again. The filesystem is used to control how data is stored on the device or provided in a virtual way by network or other services.

The standard form of the **mount** command is:

mount -t type device dir

This tells the kernel to attach the filesystem found on device (which is of type type) at the

Manual page mount(8) line 1 (press h for help or q to quit)

```
Fsck(8)                                     System Administration                                     Fsck(8)

NAME
    fsck - check and repair a Linux filesystem

SYNOPSIS
    fsck [-lsAVRTMNP] [-r [fd]] [-C [fd]] [-t fstype] [filesystem...] [--] [fs-specific-options]

DESCRIPTION
    fsck is used to check and optionally repair one or more Linux filesystems. filesystem can be a
    device name (e.g., /dev/hdc1, /dev/sdb2), a mount point (e.g., /, /usr, /home), or a filesystem
    label or UUID specifier (e.g., UUID=8868abf6-88c5-4a83-98b8-bfc24057f7bd or LABEL=root).
    Normally, the fsck program will try to handle filesystems on different physical disk drives in
    parallel to reduce the total amount of time needed to check all of them.

    If no filesystems are specified on the command line, and the -A option is not specified, fsck
    will default to checking filesystems in /etc/fstab serially. This is equivalent to the -As
    options.

    The exit status returned by fsck is the sum of the following conditions:

    0
        No errors

    1
        Filesystem errors corrected

    2
        System should be rebooted

    4
        Filesystem errors left uncorrected

Manual page fsck(8) line 1 (press h for help or q to quit)
```

Рисунок 9: Команда fsck

```

MKFS(8)                                     System Administration      MKFS(8)

NAME
    mkfs - build a Linux filesystem

SYNOPSIS
    mkfs [options] [-t type] [fs-options] device [size]

DESCRIPTION
    This mkfs frontend is deprecated in favour of filesystem specific mkfs.<type> utils.

    mkfs is used to build a Linux filesystem on a device, usually a hard disk partition. The device argument is either the device name (e.g., /dev/hda1, /dev/sdb2), or a regular file that shall contain the filesystem. The size argument is the number of blocks to be used for the filesystem.

    The exit status returned by mkfs is 0 on success and 1 on failure.

    In actuality, mkfs is simply a front-end for the various filesystem builders (mkfs.fstype) available under Linux. The filesystem-specific builder is searched for via your PATH environment setting only. Please see the filesystem-specific builder manual pages for further details.
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OPTIONS
    -t, --type type
        Specify the type of filesystem to be built. If not specified, the default filesystem type (currently ext2) is used.

    fs-options
        Filesystem-specific options to be passed to the real filesystem builder.

    -V, --verbose

Manual page mkfs(8) line 1 (press h for help or q to quit)

```

Рисунок 10: Команда mkfs

```
KILL(1)                                     User Commands                                KILL(1)

NAME
    kill - terminate a process

SYNOPSIS
    kill [-signal|-s signal|-p] [-q value] [-a] [--timeout milliseconds signal] [--] pid|name...

    kill -l [number|0xsigmask] | -L

    kill -d pid

DESCRIPTION
    The command kill sends the specified signal to the specified processes or process groups.

    If no signal is specified, the TERM signal is sent. The default action for this signal is to terminate the process. This signal should be used in preference to the KILL signal (number 9), since a process may install a handler for the TERM signal in order to perform clean-up steps before terminating in an orderly fashion. If a process does not terminate after a TERM signal has been sent, then the KILL signal may be used; be aware that the latter signal cannot be caught, and so does not give the target process the opportunity to perform any clean-up before terminating.

    Most modern shells have a builtin kill command, with a usage rather similar to that of the command described here. The --all, --pid, and --queue options, and the possibility to specify processes by command name, are local extensions.

    If signal is 0, then no actual signal is sent, but error checking is still performed.

ARGUMENTS
    The list of processes to be signaled can be a mixture of names and PIDs.

Manual page kill(1) line 1 (press h for help or q to quit)
```

3. Выводы по проделанной работе

В ходе данной работы мы ознакомились с файловой системой Linux, её структурой, именами и содержанием каталогов. Научились совершать базовые операции с файлами, управлять правами их доступа для пользователя и групп. Ознакомились с Анализом файловой системы. А также получили базовые навыки по проверке использования диска и обслуживанию файловой системы.